# Assignment 1： 上吊的小人

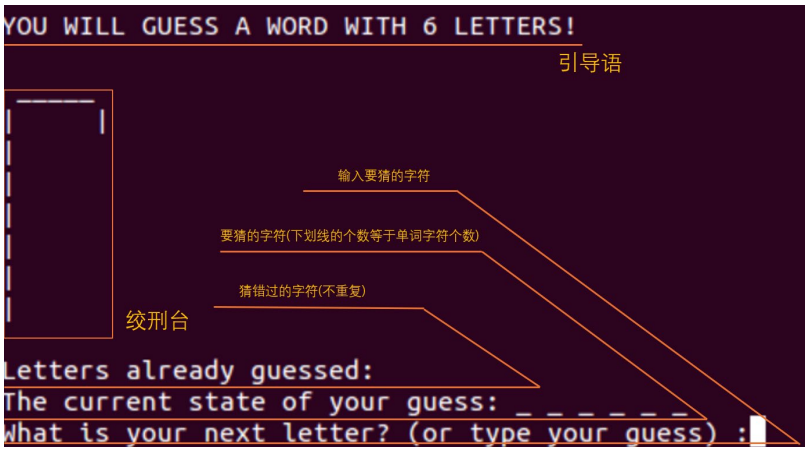
截止日期: 2025.11.20

- 作业需要提交两份文件
  - 代码源文件(.java)
  - 运行结果展示(用pdf文档)

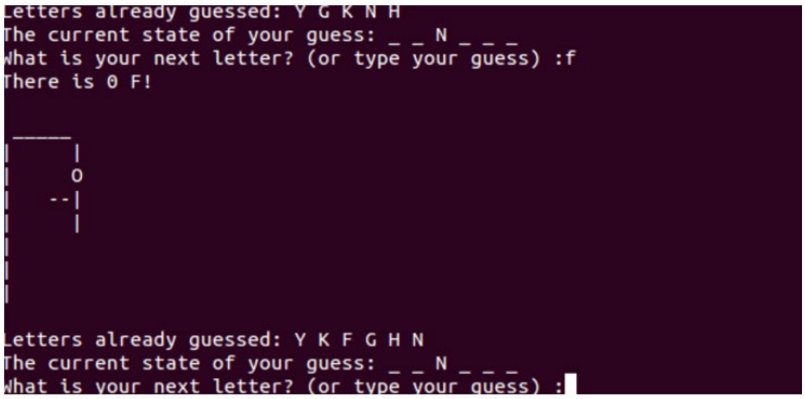
## 程序要求

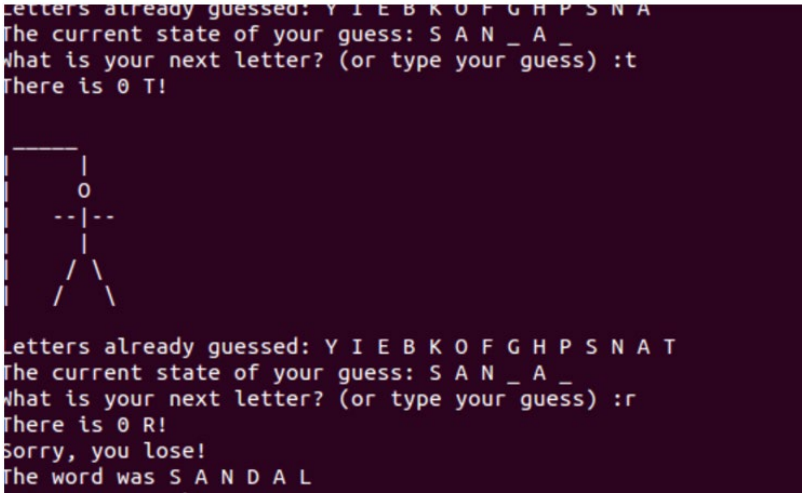
这是一个Java编写的猜单词游戏，该游戏由建立词库和猜单词两个部分来组成。

1. 建立词库
   A. 开始游戏前，我们需要准备一个英文语句作为输入。
   B. 游戏开始的时候会首先分析语句中所包含的所有英文单词，用来建立词库。
   C. 词库的要求:
      i. 语句中出现的单词要都在，但不能有重复的单词存在
      ii. 所有的单词是小写字母，不能包含标点符号。
      iii. 数字和标点符号不能包含在词库当中。
2. 猜单词游戏
   A. 游戏开始会显示欢迎界面，包含内容如下:
      i. 欢迎语句和游戏介绍以及简单的玩法向导
      ii. 菜单
         1. 查看词库
         2. 开始游戏
         3. 退出游戏
   B. 查看词库：列出词库内所有的单词。
   C. 退出游戏：结束游戏
   D. 开始游戏
      i. 系统会随机选取一个单词作为目标单词，玩家看不到目标单词。
      ii. 游戏界面




iii. 猜对一个字符，在对应的下划线上显示字符，猜错一个字符，绞刑台上的小人会完成一步上吊




iv. 在小人完全上吊之前猜对了所有字符则胜利，否则失败，均返回欢迎界面，可选择菜单。

以下为建立词库的代码运行截图：

```
Please enter a sentence ending with . or ! or ?:
what is your name
输入非法或缺少句末符，请重新输入。
Please enter a sentence ending with . or ! or ?:
what is your name 21321312321
输入非法或缺少句末符，请重新输入。
Please enter a sentence ending with . or ! or ?:
what is your 姓名.
输入非法或缺少句末符，请重新输入。
Please enter a sentence ending with . or ! or ?:

句子不能为空，请重新输入。
Please enter a sentence ending with . or ! or ?:
123 123
输入非法或缺少句末符，请重新输入。
Please enter a sentence ending with . or ! or ?:
123 123.
句中必须包含至少一个字母，请重新输入。
Please enter a sentence ending with . or ! or ?:
what is your name?
```

以下为猜单词游戏的欢迎界面：

```
Welcome to the “Guess the Word” game! Ready to test your vocabulary and luck?
The game introductions are following:
    The system will randomly choose a hidden English word.
    You can’t see the word.
    You guess letters to reveal the word.
    Each wrong guess adds one step to the hanging figure.
    If you find all letters before the figure is fully hanged, you win; otherwise you lose.
The simple How-to-Play guides are following:
    1. Enter a letter to start guessing.
    2. If the letter is correct, it appears in the word.
    3. If it’s wrong, the hangman gets closer to being hanged.
    4. Finish the word before the hangman is complete.
    5. Good luck!
==================================
                Menu
==================================
1. View the wordbook
2. Start the game
3. Exit the game
Please enter your choice: |
```

以下为猜单词游戏的查看词库界面：

```
==================================
                Menu
==================================
1. View the wordbook
2. Start the game
3. Exit the game
Please enter your choice: 1
The wordbook: is, name, what, your
Please press any key to return:
```

以下为猜单词游戏的开始游戏界面中，胜利时的截图：

```
YOU WILL GUESS A WORD WITH 2 LETTERS!

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|     |
|
|
|
|
|

Letters already guessed:
The current state of your guess: _ _
What is your next letter? (or type your guess) :i
There is 1 I!
```

```
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|     |
|
|
|
|
|

Letters already guessed: I
The current state of your guess: i _
What is your next letter? (or type your guess) :w
There is 0 W!
```

```
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|     |
|     o
|
|
|
|

Letters already guessed: I W
The current state of your guess: i _
What is your next letter? (or type your guess) :w
You have already guessed the letter!
What is your next letter? (or type your guess) :dasdasd
Your letter is invalid!
What is your next letter? (or type your guess) :123123fasssfdas asd
Your letter is invalid!
What is your next letter? (or type your guess) :s
There is 1 S!

Congratulations, you win!
The word was I S
```

```
Congratulations, you win!
The word was I S
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    You guess letters to reveal the word.
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    2. If the letter is correct, it appears in the word.
    3. If it’s wrong, the hangman gets closer to being hanged.
    4. Finish the word before the hangman is complete.
    5. Good luck!
===================================
                Menu
===================================
1. View the wordbook
2. Start the game
3. Exit the game
Please enter your choice:
```

以下为猜单词游戏的开始游戏界面中，失败时的截图：

```
Please enter your choice: 2
YOU WILL GUESS A WORD WITH 4 LETTERS!


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|     |
|
|
|
|
|

Letters already guessed:
The current state of your guess: _ _ _ _
What is your next letter? (or type your guess) :z
There is 0 Z!
```

```
 -----
|     |
|     o
|
|
|
|

Letters already guessed: Z
The current state of your guess: _ _ _ _
What is your next letter? (or type your guess) :a
There is 0 A!
```

```
 -----
|     |
|     o
|     |
|
|
|

Letters already guessed: Z A
The current state of your guess: _ _ _ _
What is your next letter? (or type your guess) :q
There is 0 Q!
```

```
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|     |
|     o
|     |
|     |
|
|

Letters already guessed: Z A Q
The current state of your guess: _ _ _ _
What is your next letter? (or type your guess) :f
There is 0 F!
```

```
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|     |
|     o
|    -|
|     |
|
|

Letters already guessed: Z A Q F
The current state of your guess: _ _ _ _
What is your next letter? (or type your guess) :b
There is 0 B!
```

```
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|     |
|     o
|   --|
|     |
|
|

Letters already guessed: Z A Q F B
The current state of your guess: _ _ _ _
What is your next letter? (or type your guess) :c
There is 0 C!
```

```
 -----
|     |
|     o
|   --|-
|     |
|
|

Letters already guessed: Z A Q F B C
The current state of your guess: _ _ _ _
What is your next letter? (or type your guess) :w
There is 0 W!
```

```
 -----
|     |
|     o
|   --|--
|     |
|
|

Letters already guessed: Z A Q F B C W
The current state of your guess: _ _ _ _
What is your next letter? (or type your guess) :u
There is 1 U!
```

```
 -----
|     |
|     o
|   --|--
|     |
|
|

Letters already guessed: Z A Q F B C W U
The current state of your guess: _ _ u _
What is your next letter? (or type your guess) :g
There is 0 G!
```

```
 -----
|     |
|     o
|   --|--
|     |
|    /
|

Letters already guessed: Z A Q F B C W U G
The current state of your guess: _ _ u _
What is your next letter? (or type your guess) :l
There is 0 L!
```

```
 -----
|     |
|     o
|   --|--
|     |
|    /
|   /

Letters already guessed: Z A Q F B C W U G L
The current state of your guess: _ _ u _
What is your next letter? (or type your guess) :p
There is 0 P!
```

```
 -----
|     |
|     o
|   --|--
|     |
|    / \
|   /

Letters already guessed: Z A Q F B C W U G L P
The current state of your guess: _ _ u _
What is your next letter? (or type your guess) :t
There is 0 T!
```

```
 -----
|     |
|     o
|   --|--
|     |
|    / \
|   /   \

Letters already guessed: Z A Q F B C W U G L P T
The current state of your guess: _ _ u _
What is your next letter? (or type your guess) :r
There is 1 R!
```

```
 -----
|     |
|     o
|   --|--
|     |
|    / \
|   /   \

Letters already guessed: Z A Q F B C W U G L P T R
The current state of your guess: _ _ u r
What is your next letter? (or type your guess) :b
You have already guessed the letter!
What is your next letter? (or type your guess) :n
There is 0 N!
```

```
Sorry, you lose!
The word was Y O U R
Welcome to the “Guess the Word” game! Ready to test your vocabulary and luck?
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    4. Finish the word before the hangman is complete.
    5. Good luck!
===================================
                Menu
===================================
1. View the wordbook
2. Start the game
3. Exit the game
Please enter your choice:
```

以下为猜单词游戏的退出游戏界面：

```
Welcome to the “Guess the Word” game! Ready to test your vocabulary and luck?
The game introductions are following:
    The system will randomly choose a hidden English word.
    You can’t see the word.
    You guess letters to reveal the word.
    Each wrong guess adds one step to the hanging figure.
    If you find all letters before the figure is fully hanged, you win; otherwise you lose.
The simple How-to-Play guides are following:
    1. Enter a letter to start guessing.
    2. If the letter is correct, it appears in the word.
    3. If it’s wrong, the hangman gets closer to being hanged.
    4. Finish the word before the hangman is complete.
    5. Good luck!
===================================
                Menu
===================================
1. View the wordbook
2. Start the game
3. Exit the game
Please enter your choice: 3

进程已结束，退出代码为 0
```